

Development of Pipeline Inspection Robot for Detection of Pipeline DefectsSavitha A C¹, Jayanth Nayaka R², Madhushankar S³, Manoj Gowda H D⁴,Shashank D J⁵, Madhu kumar K M^{*6}Assistant Professor¹, Student^{1,3,4,5}, Industry Professional⁶Department of Electronics & Communication Engineering^{1,2,3,4,5}JSS Academy of Technical Education, Bangalore, India^{1,2,3,4,5}L & T Technology Service, Bengaluru, India⁶**Abstract**

Pipelines are crucial for the transportation of fluids in various sectors, including oil and gas, water supply, and sewage systems. Regular inspection is essential for maintaining pipeline integrity and preventing catastrophic failures. Traditional inspection methods often involve manual labor, which can be unsafe and time-consuming. The introduction of robotic technologies represents a significant advancement in this domain, allowing for efficient and safe inspections. Our work seeks to create a Pipe Inspection Robot that incorporates modern robotics and computer vision techniques. The Pipe Inspection Robot is designed to efficiently automate the inspection of pipelines to detect cracks and structural weaknesses. Utilizing an ESP32 micro controller connected to DC motors for movement, and a Python application that employs YOLOv5 for image processing, the system provides real-time identification and localization of defects. Crack detection is augmented by Light Dependent Resistors (LDR) connected to the ESP32, and distance calculations are based on the rotation of the DC motors, ensuring precise crack measurements. This innovative amalgamation of technology aims to enhance infrastructure maintenance while reducing human exposure to hazardous environments.

Keywords: Crack detection, Pipe line inspection Robot, Ultra sonic sensor, Water



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1. Introduction

Pipelines are one of the most efficient and widely used methods for transporting oil, gas, water, and industrial fluids over long distances. As these pipelines age, they are exposed to various environmental and operational conditions such as corrosion, pressure fluctuations, temperature changes, and mechanical stress. These factors can lead to defects including cracks, leaks, corrosion pits, blockages, and weld failures. If not detected early, such defects may result in severe consequences like environmental pollution, product loss, fire hazards, or even catastrophic pipeline failures. Traditional inspection methods are often manual, time-consuming, and sometimes unsafe. External inspections require digging, shutting down sections of the pipeline, or operating in hazardous areas. Internal inspection tools are effective but expensive, and many narrow or complex pipelines, especially small-diameter or curved pipelines, cannot be inspected by conventional pigging devices. This creates a strong need for a compact, adaptable, and cost effective inspection solution. To address these challenges, the development of pipeline inspection robots has emerged as a modern engineering approach. These robots are designed to travel inside pipelines, capture real time data, and identify defects using sensors such as cameras, ultrasonic sensors, infrared sensors, or magnetic flux leakage detectors. Robotics-based inspection enhances safety, reduces human effort, and allows continuous monitoring even in difficult-to-access areas.

A. Motivation

Pipelines form the backbone of many critical industries, including oil and gas, water supply, chemical processing, and manufacturing. Failure in these pipelines can lead to severe economic losses, safety hazards, and environmental damage. Many incidents in the past have shown that undetected corrosion, cracks, or blockages can result in leaks, explosions, or contamination of natural resources. Despite this importance, pipeline inspection in many industries is still done manually or with bulky equipment that cannot reach narrow, curved, or buried pipelines. Human inspection is risky, slow, and often inaccurate. Additionally, shutting down a pipeline for inspection increases operational costs. These challenges highlight a clear need for a smarter, safer, and more efficient inspection solution. The motivation behind developing a pipeline inspection robot comes from the desire to:

- Enhance safety by reducing human involvement in dangerous environments.
- Detecting defects early to prevent catastrophic failures and costly downtime.
- Inspect pipelines of various sizes and orientations where traditional inspection tools cannot operate.
- Make inspection more affordable for industries that cannot invest in expensive commercial robots.
- Promote automation and smart monitoring, aligning with modern Industry 4.0 standards.

Creating a compact, reliable, and cost-effective inspection robot will significantly improve maintenance practices and ensure long-term sustainability of pipeline infrastructures. This motivates the design and development of an innovative robotic solution capable of navigating complex pipelines and accurately detecting defects. There is a lack of cost-effective, automated, and real-time systems for detecting cracks and blockages in pipelines. The current methods are either manual, expensive, or lack remote monitoring capability. Hence, there is a need for a pipeline inspection robot with integrated sensors and IoT -based alerts to improve pipeline safety and maintenance efficiency. Pipelines are critical for transporting water, oil, gas, and sewage, but they often face issues such as cracks, leaks, blockages, and surface irregularities.

B. Objectives of the Study

The proposed work has following objectives:

- To design and develop a compact and efficient pipeline inspection robot capable of navigating through pipelines of various diameters and orientations.
- To integrate suitable sensors like camera, ultrasonic, infrared, or other defect detection sensors for identifying cracks, corrosion, blockages, and other pipeline defects.
- To ensure stable and reliable mobility of the robot inside the pipeline, including smooth movement through bends, joints, and vertical sections.
- To develop a real-time monitoring system for capturing and transmitting defect data such as images, video, or sensor readings.
- To create a user-friendly control system that allows either wired, wireless, or autonomous operation of the robot.
- To implement data processing methods for accurate defect detection and classification based on the collected sensor data.
- To evaluate the robot's performance in detecting defects under different pipeline conditions through testing and analysis.
- To provide a cost-effective alternative to traditional pipeline inspection methods, making the solution accessible to industries with limited resources.

2. Literature Review

Installation of quality pipelines for a longer duration is very important for social, economical and maintaining a good environmental system. This is the major motivation to maintain a healthy pipeline. A number of researchers have worked in this direction. Some of the researcher proposals have been discussed in this section. Ayman K et al. [1], proposed the 'Pipeline Leak Identification Emergency Robot Swarm', which is implemented with respect to industry. It has neural network technologies, Internet-of-Things, and robots. For the detection of cracks in barren water and waste pipelines. This system collects the images and records their places. Then the images are transferred to the cloud and do the analysis using a convolution neural network. CNN will check its severity level. CNN model presents the 80-90% accuracy score. Lv, Z., Dong et al. [2], has understood the difficulty of crack detection in underground water pipe lines in rural areas, he proposed a lightweight YOLOv8n crack detection in pipeline. This technique is based on the pipeline robots, which quickly move in water pipes and identify the obvious diseases of sewage pipelines. This YOLOv8n crack detection has very few parameters in comparison to YOLOv10n, YOLOv9t, YOLOv6n and YOLOv5n. The validation results show possible safety, improvement in maintenance, less cost especially required for urban sewage pipes. And also more efficient real time detection. K. Zhang et al, developed an MFL-based inspection system for detecting corrosion and wall thinning in steel pipelines. Their robot showed strong detection performance for subsurface defects and provided calibrated data for estimating defect depth. The study demonstrated the effectiveness of MFL sensors in industrial pipelines[3-5]. R. N. Al-Hannabi et al. This work integrates visual and ultrasonic sensors for hybrid defect detection. The robot uses autonomous navigation enabled by onboard processors and sensors. The system achieved high accuracy in identifying cracks, weld defects, and blockages, proving suitable for complex pipe networks[6-7]. Pshenin, V. et al, designed a crawler robot with wireless communication capabilities for sewer and water pipelines. Their design includes a high-resolution camera and LED illumination for visual defect detection. The robot demonstrated stable traversal in long horizontal and inclined pipelines[8-11]. Nguyen, T.L et al, [12] introduces a compact robot capable of inspecting small-diameter pipes below 50 mm using a micro-camera and IMU sensors. It highlights challenges in navigating tight bends and presents a lightweight solution suitable for residential and industrial pipelines. In Tang, R et al, [13], the robot uses guided ultrasonic waves for detecting long-range defects such as corrosion and cracks. The system can inspect long pipe sections from a single location. The authors report high sensitivity to corrosion and crack depth measurement.

3. Proposed System

The development of the pipeline inspection robot follows a systematic methodology that includes problem analysis, design, development, integration of components, and performance evaluation. The overall methodology is described in table 1.

Table 1: Pipeline Inspection Methodologies

Functional Requirements	- Robots must move inside pipelines smoothly. - Must detect defects like cracks, corrosion, and blockages. - Should capture and transmit real-time video or sensor data. - Must allow wired/wireless control. - Should operate on a reliable power supply.
Non-Functional Requirements	- High accuracy in defect detection. - Compact, lightweight, and durable design. - Safe operation without damaging the pipeline. - Cost-effective and reliable performance.
System Design	- Mechanical Structure - Sensor Integration - Control Unit - Power System - Communication Module - Software System
Hardware Assembly	- ESP32 Microcontroller – Data processing - Wireless communication - LED & LDR Sensors – Crack detection using optical method - Vibration Sensor – Blockage/uneven surface detection - Motor & Chassis – Robot movement inside pipeline - Rechargeable Battery Pack – Power supply - Wheels/Tracks – Navigation inside pipelines
Software Development	- Arduino IDE / ESP-IDF – Programming the ESP32 - Telegram Bot API – Real-time notification system - Embedded C / Micro Python – Firmware development - Serial Monitor / IoT Dashboard – Debugging & monitoring sensor data

The circuit working flow of the Disaster Site Monitoring and Human Retrieval Robot explains how each electronic subsystem interacts to perform monitoring, navigation, detection, and rescue operations in a hazardous environment. The entire system is centered around the ESP32 micro controller, which coordinates sensors, actuators, the camera module, wireless communication, and the robotic arm. Figure 1. shows the block diagram of the pipeline inspection robot. The pipeline investigation robot has mainly 3 sections: an autonomous mobile robot, a crack detector and corrosion detector and lastly alert message scheme. Along with this it has a pre- processed module for managing programming and controlling (robot application). The robotic unit is placed in work place actions are performed through the sensor unit and the response is given to LCD. Figure 2 shows the developed final design of the pipe inspection robot. The robot has an ultrasonic sensor to detect cracks in the pipeline. A camera is connected to the robot for recording and the result is displayed through LCD.

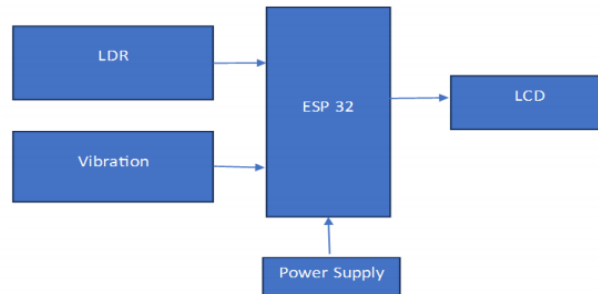


Fig.1: Block diagram of the proposed system

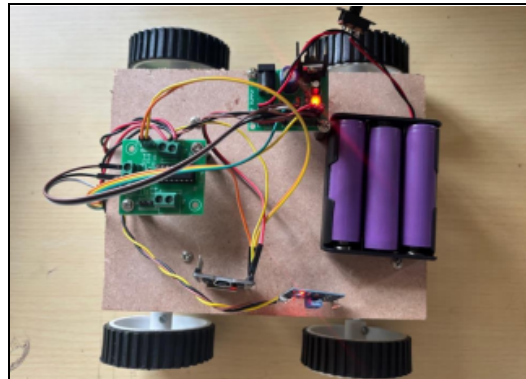


Fig.2:Final design of in pipe inspection robot

4. Results and discussion

A Pipeline Inspection Robot contains electronic systems that work together to power the robot, sense the environment, move through the pipe, and communicate information to the operator. The entire operation depends on how each circuit block interacts with the others. Our proposed system does the routine inspections along with the safety measures. System data gives robot functionalities reports and helps the analysis of inspection findings. This conceptualization framework guarantees the robot's 'reliability' and 'efficiency' in finding and addressing pipeline leaks and cracks.

A. Benefits

- Enables Inspection in Hazardous Environments
- Works in Narrow and Inaccessible Pipelines
- Real-Time Monitoring and Reporting
- High Accuracy and Efficient Fault Detection
- Reduces Operational Cost
- Improves Maintenance Planning
- Designed for Long-Distance Travel
- High Stability and Mobility in Rough Conditions
- Data Storage for Future Analysis
- Works in Both Metallic and Non-Metallic Pipelines

B. Challenges and limitations

- Communication Problems Inside Pipelines
- Limited Battery Life
- Difficulty Navigating Complex Pipeline Geometry
- Sensor Limitations
- Mechanical Wear and Tear

- Limited Payload Capacity
- Complexity of Design and Maintenance

C. Future Scope

- Better and Longer-Range Communication
- Enhanced Battery Technology
- Advanced Autonomous Navigation

Conclusion

The proposed system offers a low-cost, real-time, and automated pipeline inspection solution. By using simple yet effective sensors like LED-LDR and vibration detectors integrated with ESP32. The system ensures reliable crack and blockage detection. The inclusion of Telegram alerts enhances response time, leading to timely maintenance and reduced operational downtime. The work can be further improved with advanced sensors and AI integration for predictive maintenance. The Pipeline Inspection Robot system successfully demonstrates an efficient, safe, and modern method for inspecting pipelines. By integrating sensors, motorized mobility, communication modules, and onboard data processing, the robot is able to detect cracks, leaks, corrosion, blockages, and structural defects with high accuracy. The system improves inspection quality through real-time monitoring and reliable data collection, allowing industries to perform preventive maintenance and reduce operational downtime. Although challenges such as communication limits, battery constraints, and sensor performance in harsh environments still exist, the robot offers a strong foundation for future advancements. Overall, our work highlights how robotic automation can significantly enhance pipeline safety, reduce human risk, and increase inspection efficiency.

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